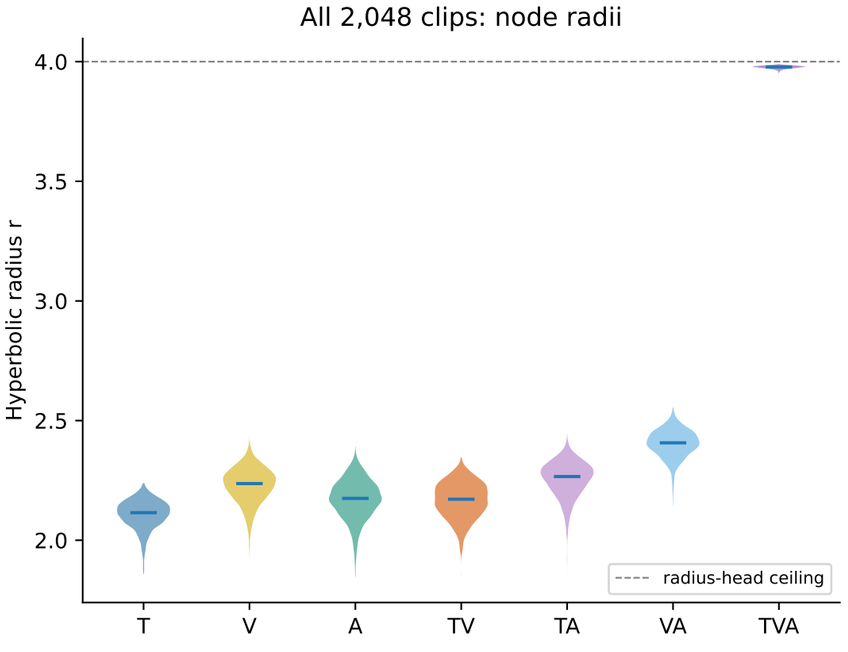
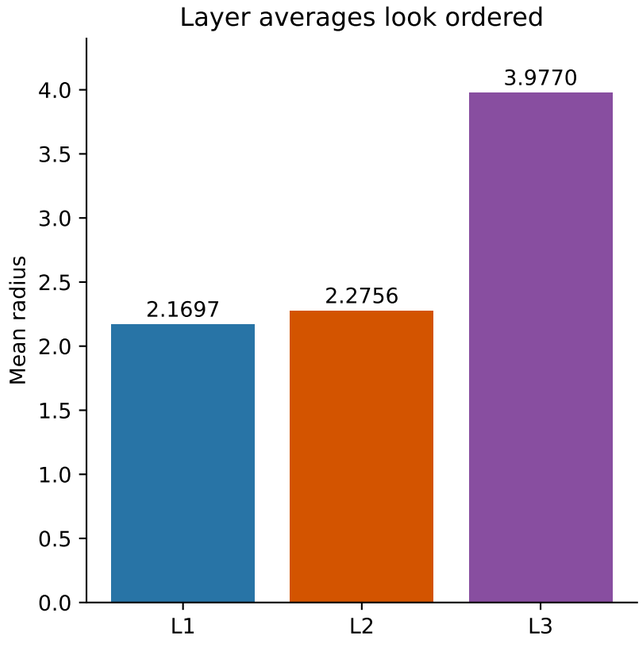
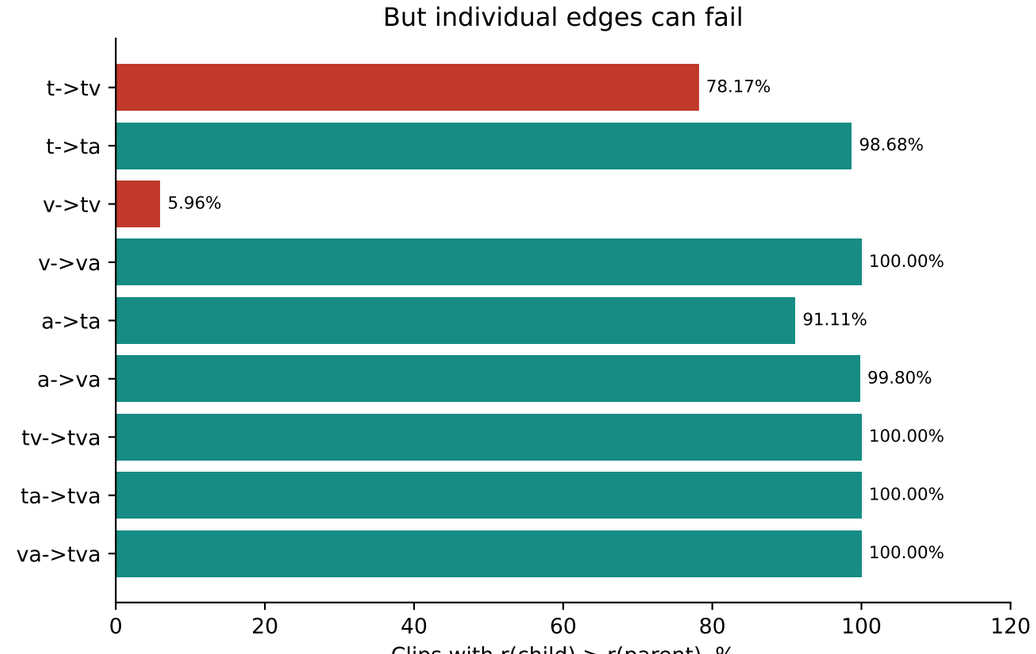
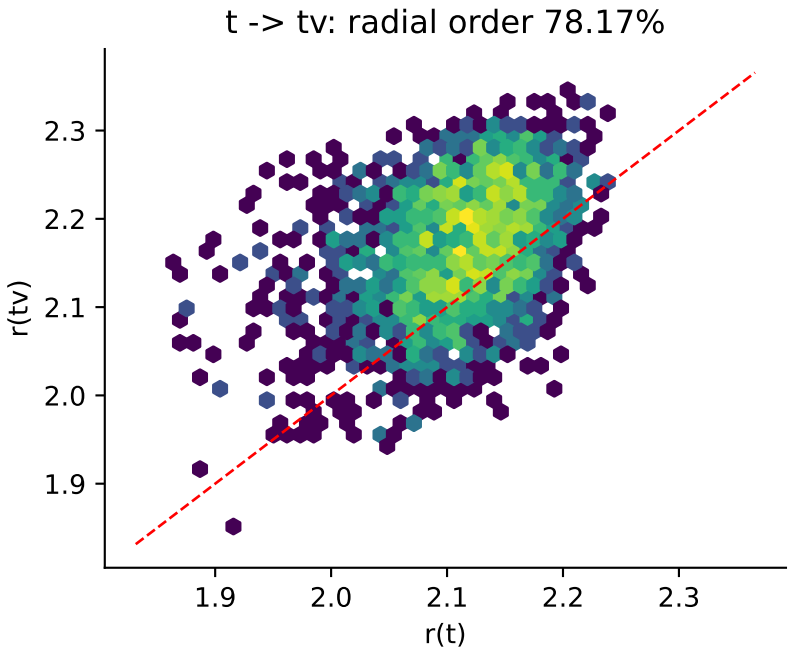
Reused C0 independent-radius trunk | epoch 30 | VAST validation | 512D, c=1

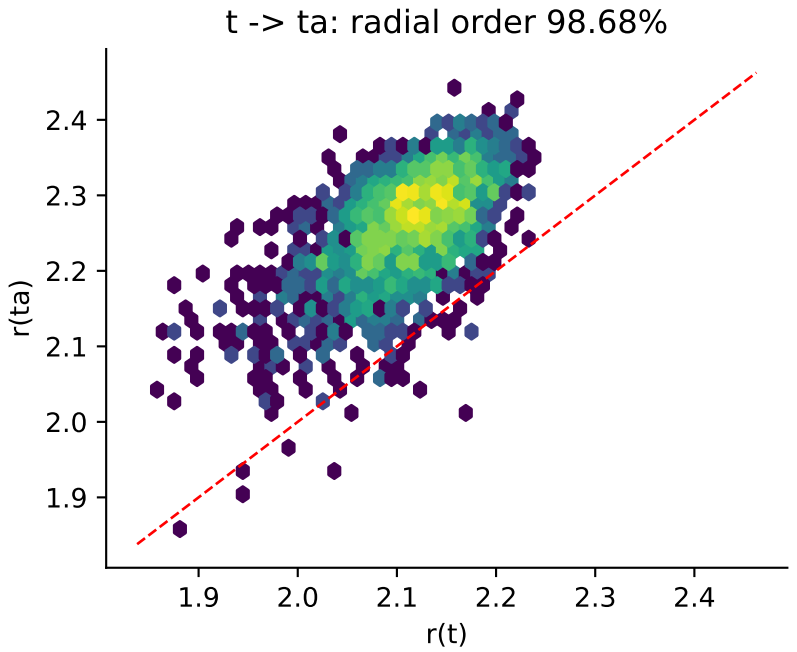


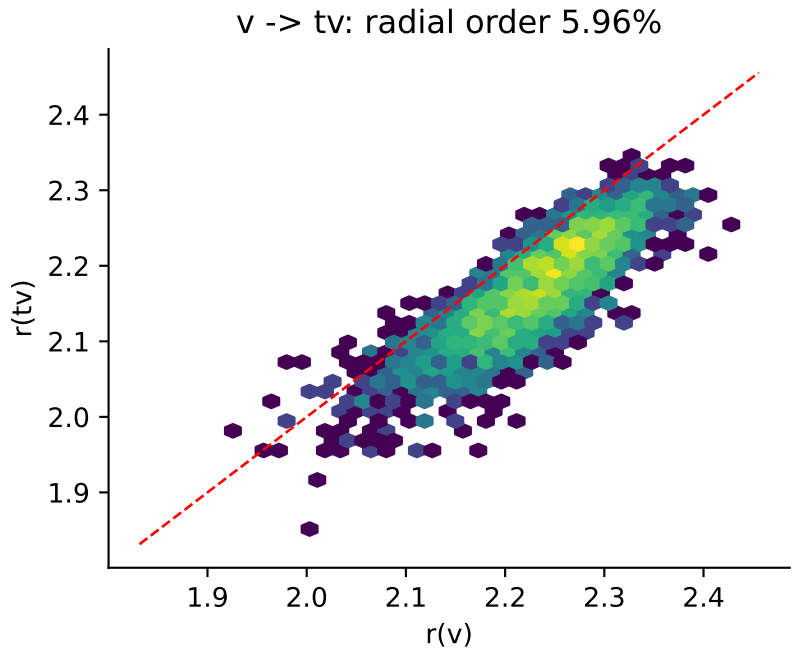


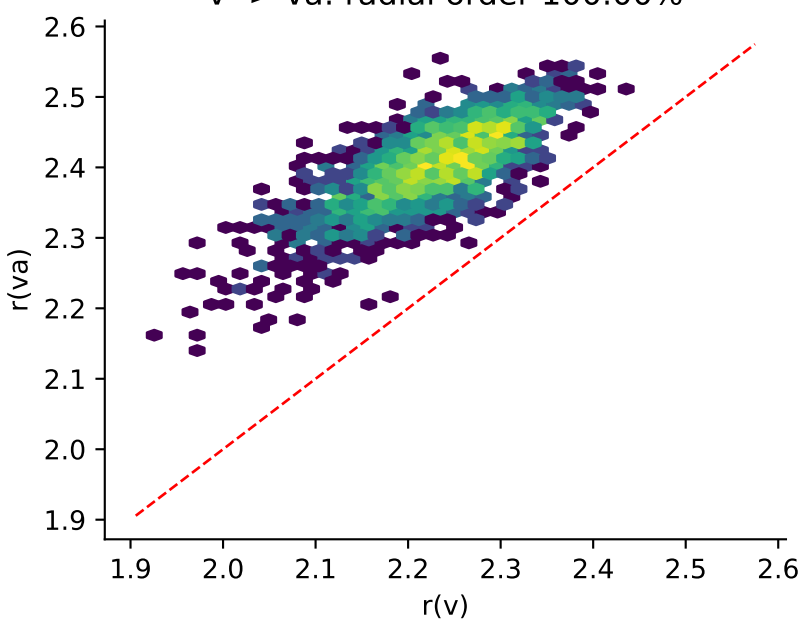


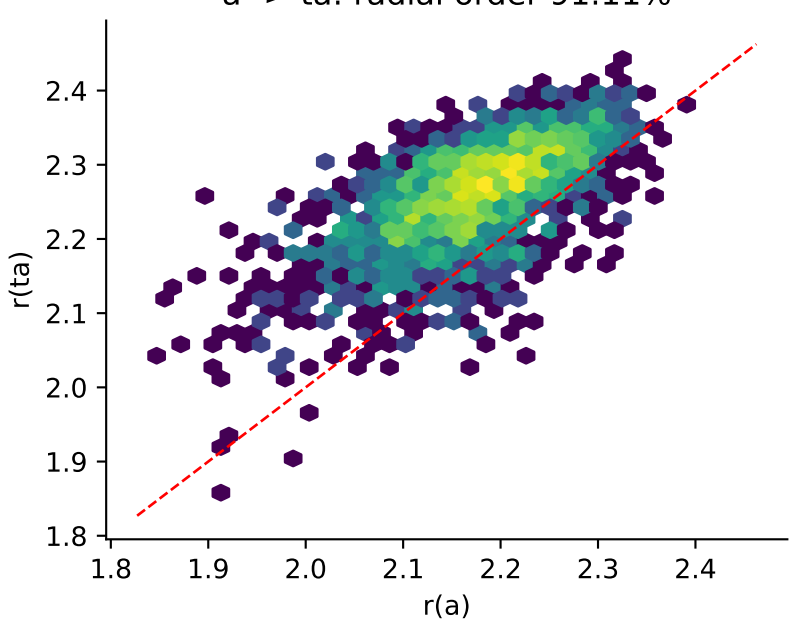
Every point is one same-clip edge; above dashed line = radially ordered

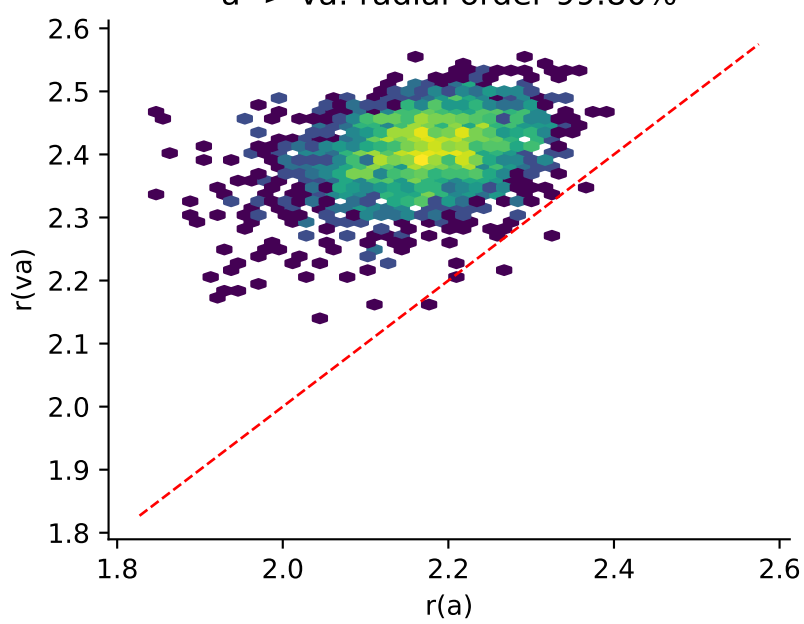


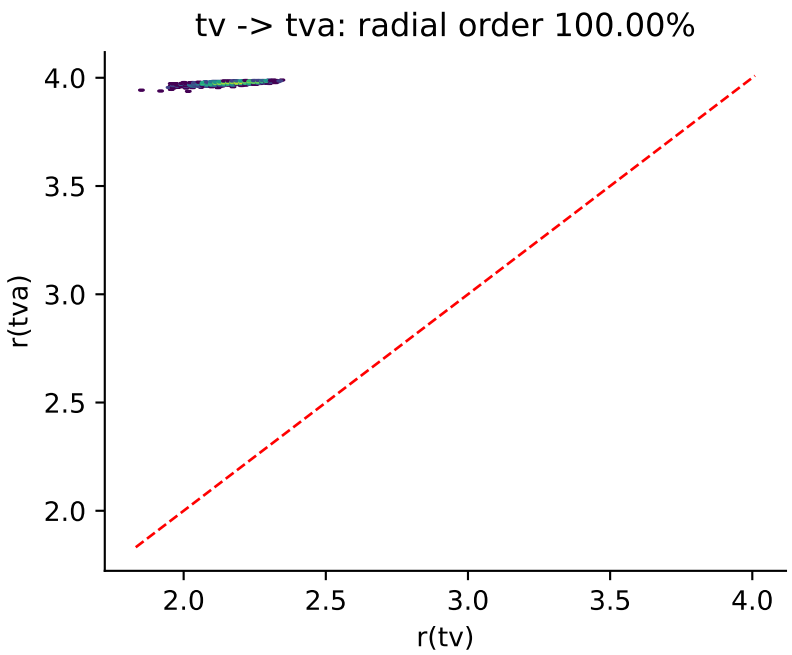


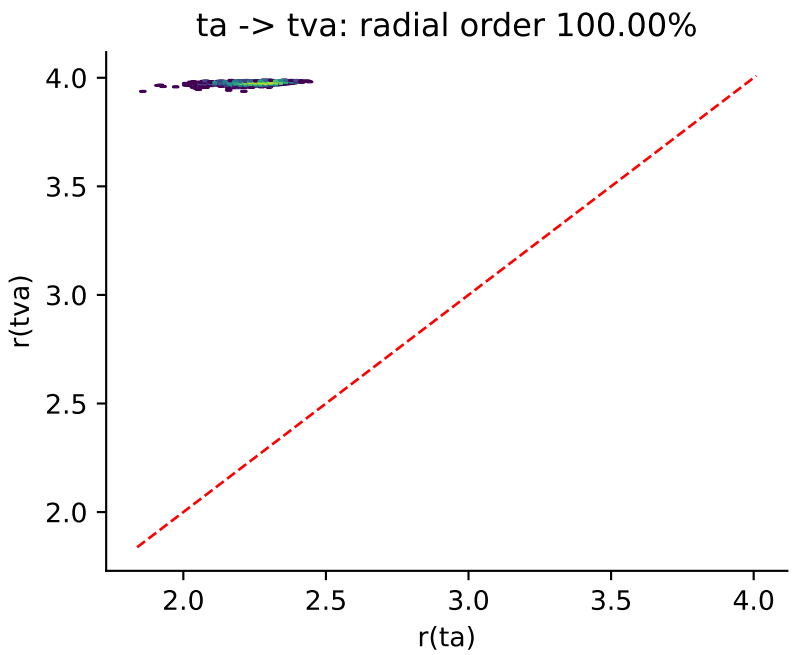


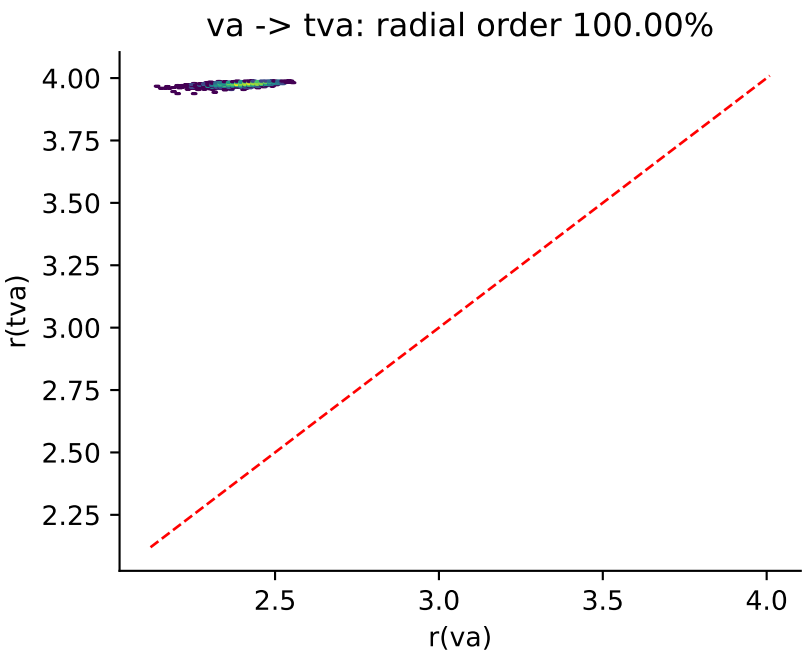


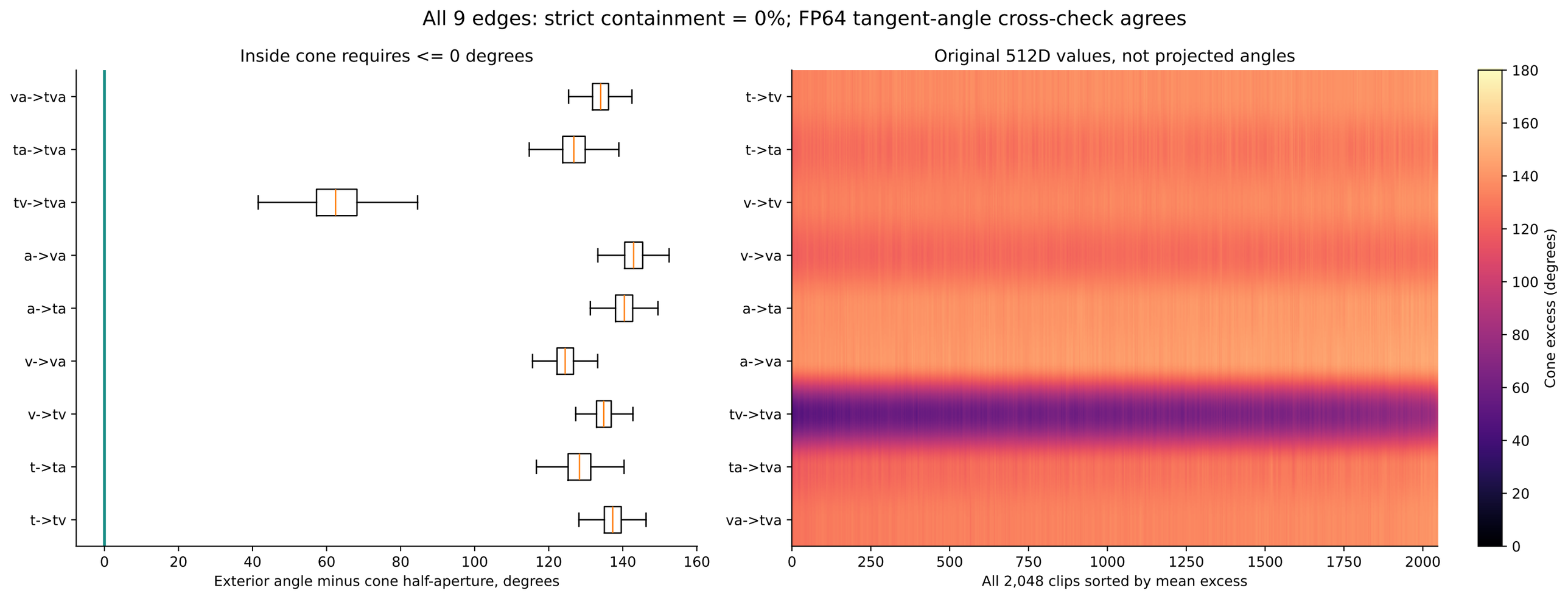




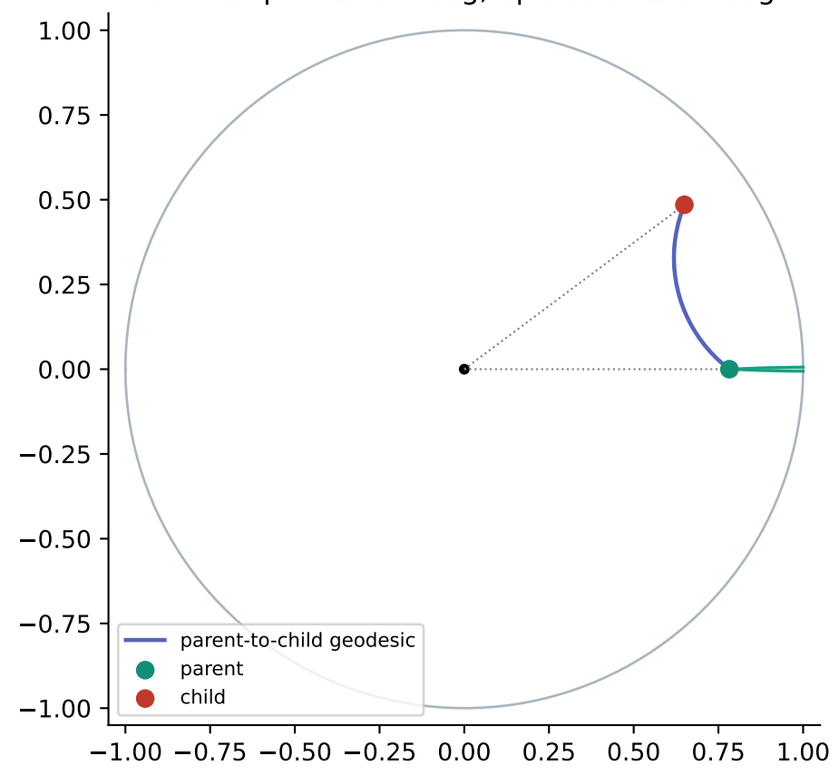


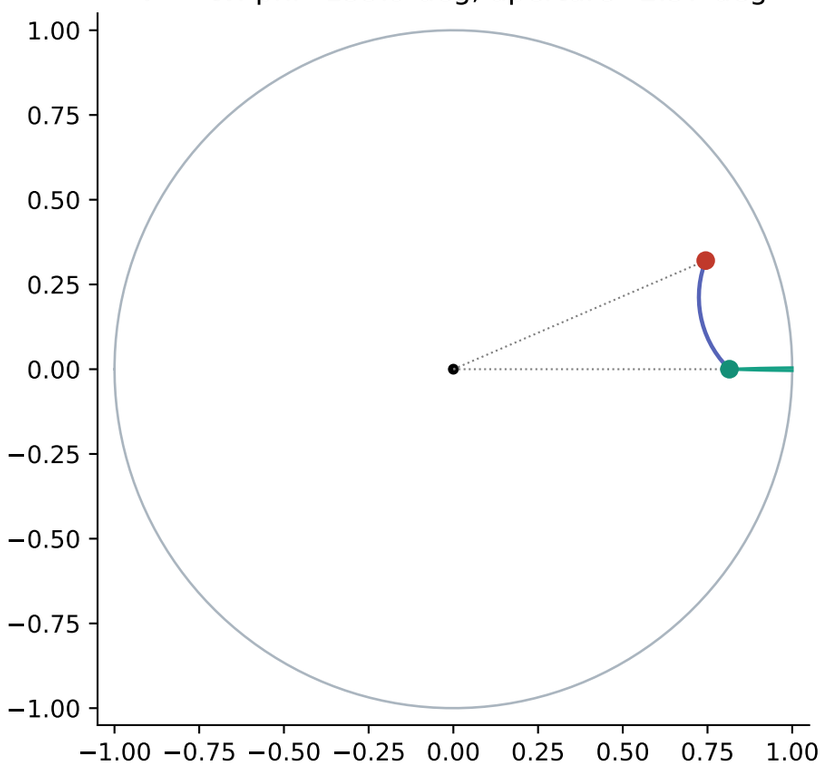


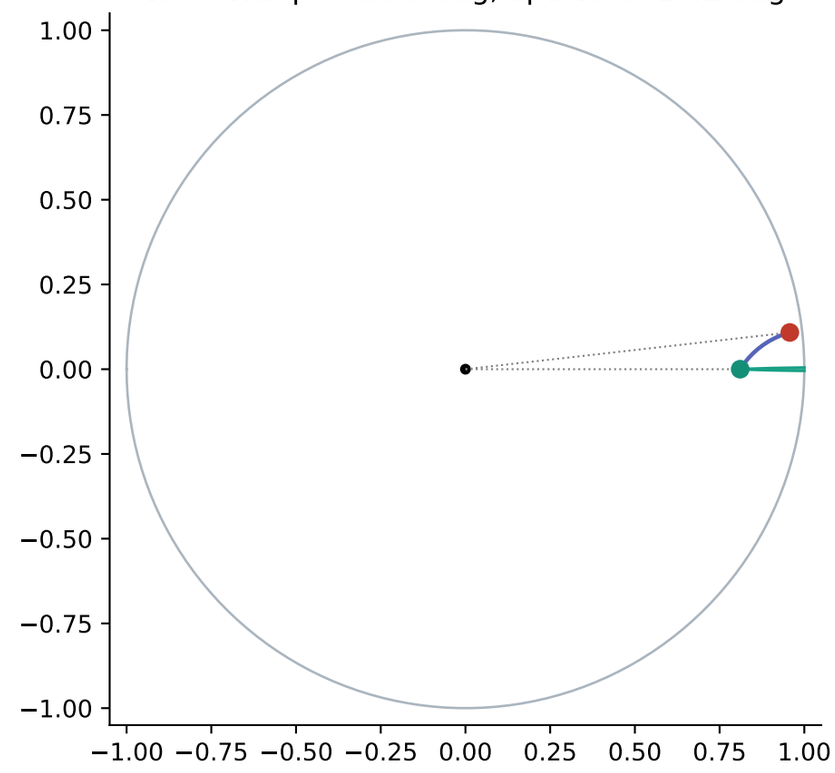


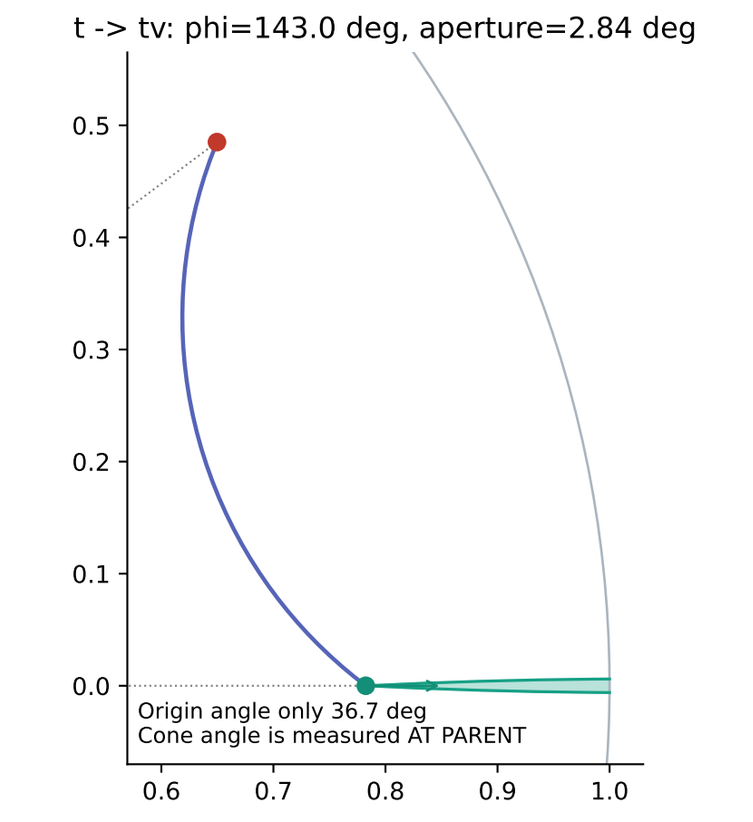

All 9 edges: strict containment = 0%; FP64 tangent-angle cross-check agrees
Inside cone requires <= 0 degrees
va->tva
ta->tva
tv->tva
a->va
a->ta
v->va
v->tv
t->ta
t->tv
0
20
40
60
80
100
120
140
160
Exterior angle minus cone half-aperture, degrees
Original 512D values, not projected angles
t->tv
t->ta
v->tv
v->va
a->ta
a->va
tv->tva
ta->tva
va->tva
0
250
500
750
1000
1250
1500
1750
2000
All 2,048 clips sorted by mean excess
Cone excess (degrees)
180
160
140
120
100
80
60
40
20
0

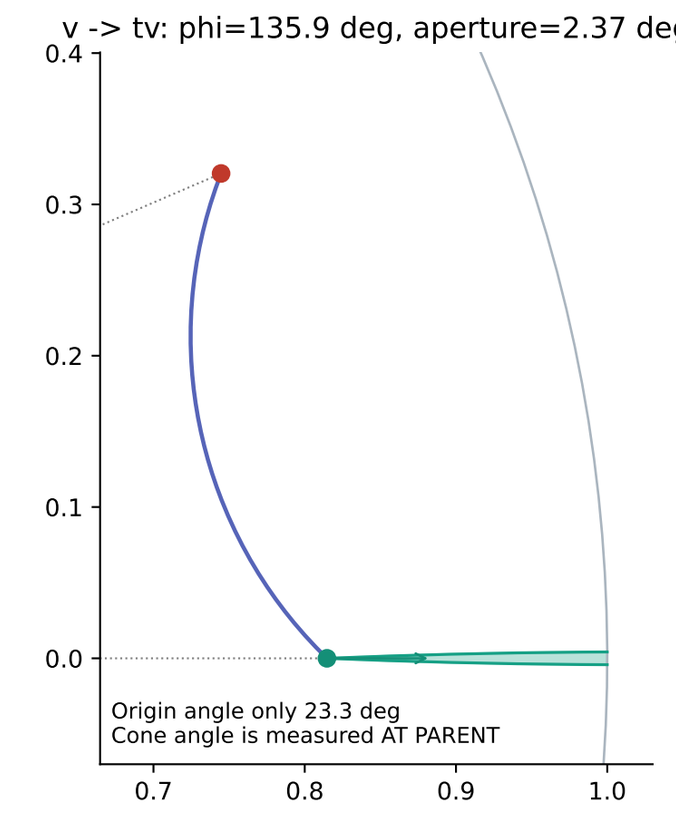
Exact pair-plane Poincare views, same representative clip row 670
Top: whole disk. Bottom: zoom. Each plane preserves O/parent/child geometry; no PCA or t-SNE.

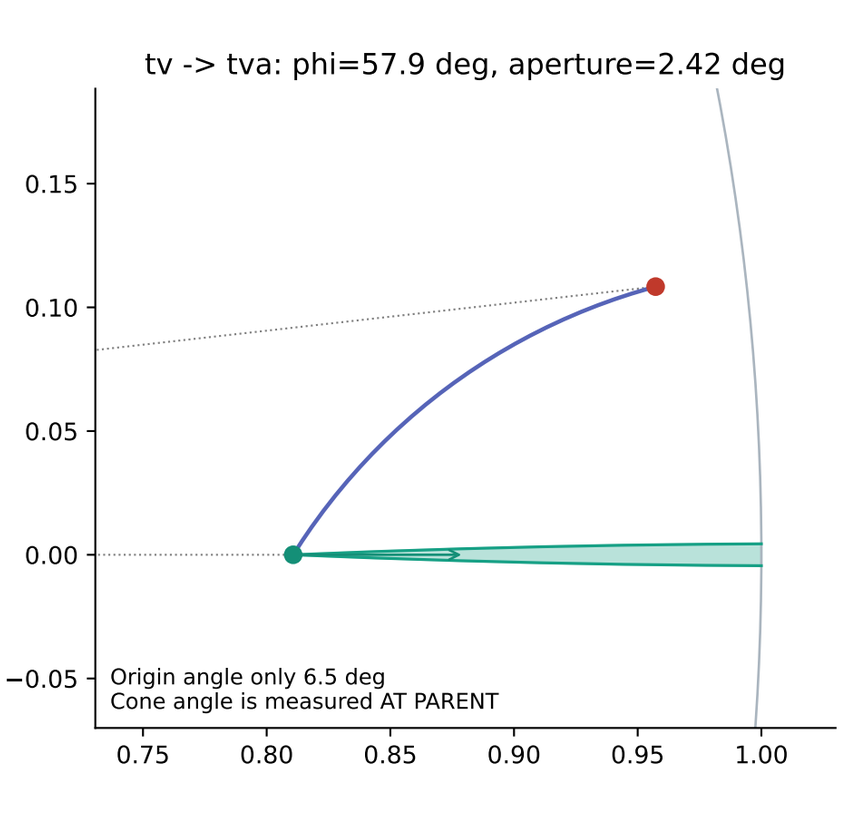












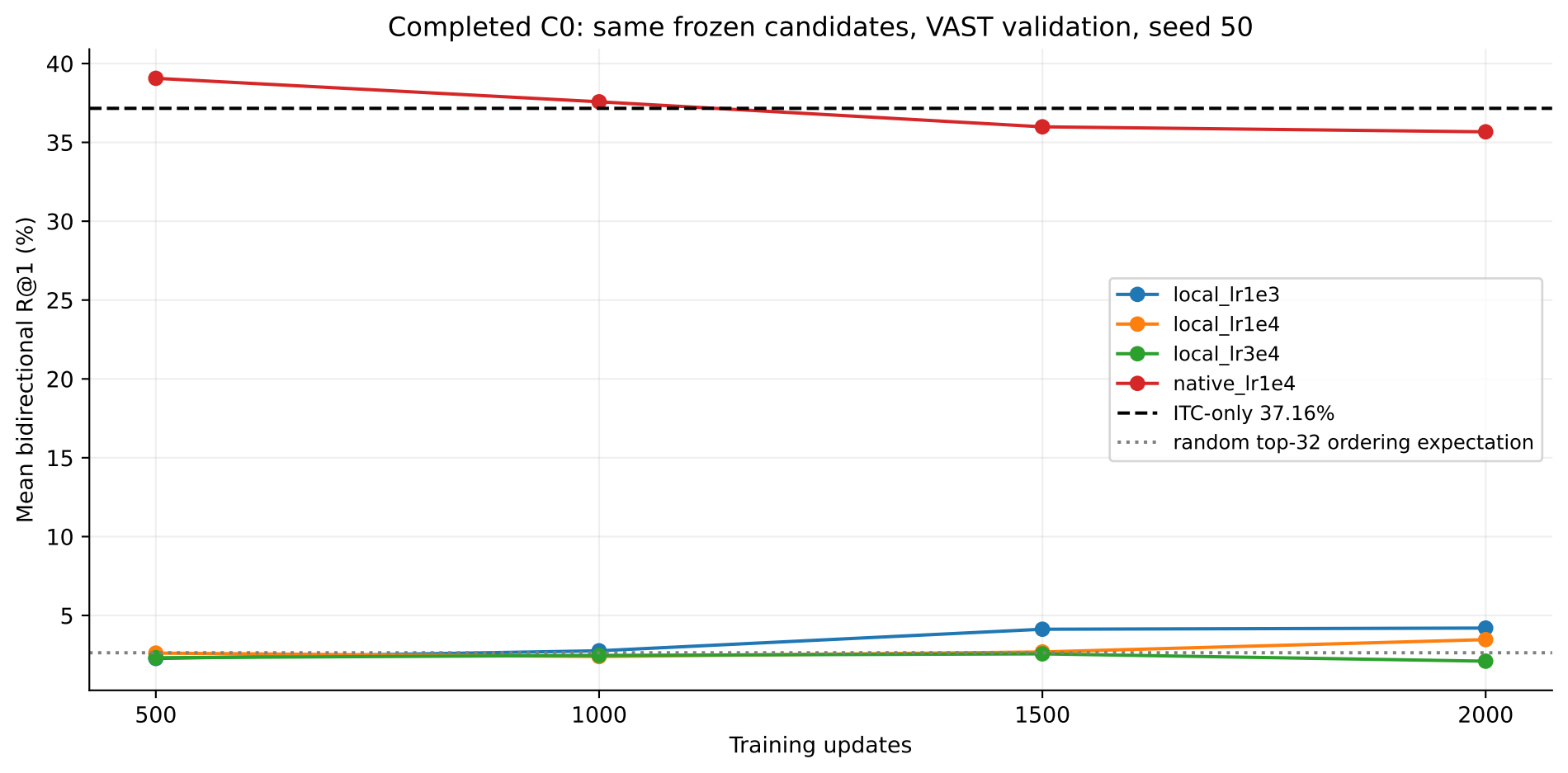
Previously saved October 26 sweep: 17 configurations, epoch 30, same VAST validation
Radial ordering can be strong while entailment-cone containment is absent.

Radial order per edge (%)

	t->tv	t->ta	v->tv	v->va	a->ta	a->va	tv->tva	ta->tva	va->tva
C0	78	99	6	100	91	100	100	100	100
M00_g05	100	100	100	100	100	100	100	100	100
M00_g1	100	100	100	100	100	100	100	100	100
M00_g2	83	99	100	100	100	100	100	100	100
M00_g4	15	60	99	100	100	100	100	100	100
M01_g05	100	100	100	100	100	100	100	100	100
M01_g1	100	100	100	100	100	100	100	100	100
M01_g2	93	100	100	100	100	100	100	100	100
M01_g4	15	54	100	100	100	100	100	100	100
M10_g05	100	100	100	100	100	100	100	100	100
M10_g1	100	100	100	100	100	100	100	100	100
M10_g2	92	100	100	100	100	100	100	100	100
M10_g4	34	83	100	100	100	100	100	100	100
M11_g05	100	100	100	100	100	100	100	100	100
M11_g1	100	100	100	100	100	100	100	100	100
M11_g2	96	100	100	100	100	100	100	100	100
M11_g4	40	94	100	100	100	100	100	100	100

Strict cone containment per edge (%)

	t->tv	t->ta	v->tv	v->va	a->ta	a->va	tv->tva	ta->tva	va->tva
C0	0	0	0	0	0	0	0	0	0
M00_g05	0	0	0	0	0	0	0	0	0
M00_g1	0	0	0	0	0	0	0	0	0
M00_g2	0	0	0	0	0	0	0	0	0
M00_g4	0	0	0	0	0	0	0	0	0
M01_g05	0	0	0	0	0	0	0	0	0
M01_g1	0	0	0	0	0	0	0	0	0
M01_g2	0	0	0	0	0	0	0	0	0
M01_g4	0	0	0	0	0	0	0	0	0
M10_g05	0	0	0	0	0	0	0	0	0
M10_g1	0	0	0	0	0	0	0	0	0
M10_g2	0	0	0	0	0	0	0	0	0
M10_g4	0	0	0	0	0	0	0	0	0
M11_g05	0	0	0	0	0	0	0	0	0
M11_g1	0	0	0	0	0	0	0	0	0
M11_g2	0	0	0	0	0	0	0	0	0
M11_g4	0	0	0	0	0	0	0	0	0


Completed C0: same frozen candidates, VAST validation, seed 50
Mean bidirectional R@1 (%)
Training updates
40
35
30
25
20
15
10
5
500
1000
1500
2000
local_lr1e3
local_lr1e4
local_lr3e4
native_lr1e4
ITC-only 37.16%
random top-32 ordering expectation